



Delivering a successful national AI strategy requires cross-economic sector collaboration as well as developing long-term capabilities in talent and business infrastructure. Image: Unsplash/Marcus Mohr

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• Unless national AI strategy is in place, it is important to track the development or implementation through owner's measures.

Various market signals show that Artificial Intelligence (AI) is a top business priority. Global corporate investment in AI increased from 2019 to 2020 by 40% to \$67.9 billion. But recently, governments' focus and spending on AI have also increased. As per an OECD study, the United States' investment in AI increased 17-fold between 2007 and 2019. Another report found AI investments in the European Union will reach €2.4 billion by 2025. In addition to AI's transformative potential across multiple industries, countries are increasingly looking at a number of AI use cases in defence and security, thereby elevating AI to the centre of discussion around a country's sovereign defence and its economic capabilities.

Countries at the forefront of AI see it as a competitive advantage in a world with fragmented yet intrinsically linked global supply chains. Developing countries see it as means to leap-frog and deliver on core development areas of their country and improve the lives of their citizens.

Increasingly, governments have published national AI strategies to create a national AI ecosystem with responsible governance. According to Stanford University's Institute for Human Centred Artificial Intelligence, 32 countries published their national AI plan, while 22 are developing one (2020).

The AI value-chain

Recent semi-conductor supply-chain challenges and geopolitical actions like the US Chips and Science Act have demonstrated that building and delivering a successful National AI strategy requires two approaches. It requires socio-economic value-chain levers, which are unpacked in the section below, as well as developing long-term capabilities across the whole value-chain from software and hardware perspective. That includes organized data, data storage, computing power, cloud systems and platforms enabling scalable AI applications.

The hardware/infrastructure part of the AI value chain remains concentrated across a handful of countries and choke points, a point well reflected in various texts. Governments will, therefore, have to develop long-term capabilities in hardware or secure their value chains through bilateral deals, trade pacts etc. The current setup of Taiwan Semiconductor Manufacturing Co. factories in the United States shows the medium to long-term view of the country's industry in strengthening its hardware and foundational capabilities in AI, even as an incumbent leader in several aspects of AI research and development and home to some of the biggest AI companies in the world.

This concentration raises pertinent questions about the gap in AI equity, reflected in numbers from a report from Deloitte and other indexes assessing national AI strategies and investments, showing that the United States, China and the United Kingdom are leaders in the space.

AI strategy: building blocks

Our published framework for developing a national AI strategy lays out critical building blocks for designing a successful strategy that protects and positively impacts citizens' lives.

First and foremost, it is important to shape a national AI ecosystem that aligns with a country's strategic priorities and strengths and weaknesses. In addition to factors highlighted in the section above, this would include foundational challenges/realities of a nation. For instance, one country facing an ageing population might leverage AI differently e.g. incentivize AI automation. Another country with a significant youth population might focus on re-skilling, up-skilling, talent development or workforce augmentation.

Similarly, a country producing a solid STEM (science, technology, engineering, mathematics) talent pool might worry less about an "AI-ready" workforce than a country with a shortage of STEM talent, which might focus on attracting external talent and rethink its education programmes. This national analysis also includes a prioritization of industry sectors. That is to say, countries with a strong manufacturing industry may prioritize AI investments in that sector versus nations that want to build out their strong agricultural sector through AI-powered innovation such as precision agriculture.

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International collaboration and multi-stakeholder collaboration is an accelerator to becoming an AI-ready nation.

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A sound AI strategy has specific, measurable, achievable, relevant, and time-bound (SMART) objectives and investment targets regarding talent, infrastructure, research and development, industry transformation, public-private collaboration and standards and regulation, including AI ethics and soft laws.

When it comes to law and regulation, a national AI strategy is ultimately an extension of a nation's digital and data strategy and associated laws. That includes existing data protection laws, privacy laws and addressing other related ethical concerns, which all build the foundation for an AI economy that works for its citizens.

Equally important is creating a strong research collaboration between academia, industry and government to boost a national AI ecosystem, similar to UK's Alan Turing Institute or Canada's CIFAR. Finally, international cooperation is critical for success due to the global interconnectedness of the AI value chain and the global distribution of capabilities and expertise.

Have you read?

- How Rwanda's vision for data and AI is revolutionizing its services and healthcare system
- Artificial intelligence - good or bad for public health?
- How does gender impact opinions about artificial intelligence?

Measuring progress

While writing a robust National AI strategy is a large undertaking, being on track with implementation and measuring progress can be challenging. The United Kingdom's national AI strategy acknowledges this and has mentioned a time-divided roadmap of its monitoring and assessment indicators to be released soon.

"Clearly, having a national AI plan is a necessary but not sufficient condition to achieve the goals of the various AI plans circulating around the world [...]", states a piece by Brookings' TechFutur blog.

Hence it is important to monitor indicators over time to track progress. Various global indices, such as the Tortoise Index or Oxford Insights Index, measuring national AI maturity and readiness, provide a good start on potential national matrices to track over time. In addition, SMART indicators and proxies for each dimension and priority area form a nation's AI strategy. Examples include:

- The number of AI-related PhDs and scientists.
- The number of AI publications and conferences.
- The number of AI patents.
- AI-related university graduates.
- Computing power.
- VC investments in AI companies.

All of these are linked to critical pillars such as research, talent, infrastructure and investments.

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Road ahead

Given the fast-paced development of AI and its application, nations need to plan, be ready and stay agile. A national AI strategy is a good start but governments need to accommodate technological advancements and changing applications. Keeping track of progress and making sure a country works towards its objectives is the most critical part of a working national AI strategy.

Agile governance here is key. An example of this is the recent and rapid rise of generative AI across sectors, disrupting labour markets in the creative fields and calling for new metrics around AI-augmented or AI-displaced skills in content and design.

International collaboration and multi-stakeholder collaboration is an accelerator to becoming an AI-ready nation. In that context, the Forum launched its National AI Strategy Peer Network last year, which puts into action its published blueprint helping governments design their national AI strategy. Its aim is for governments and experts to share best practice and learnings when designing and implementing a national AI strategy that works for its citizens.

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